Day 3: DNS Server Configuration

Name: Ratan Kumawat | Institute: TOPS Technologies

How DNS works (theory):

DNS (Domain Name System) works like a phone book of the internet. People remember names like www.musicmania.in, but computers talk using IP addresses. DNS converts the name into an IP address.

1. The user types www.musicmania.in in the browser. The computer first checks its own DNS cache and the hosts file.
2. If it is not found, the computer asks the DNS resolver (the configured DNS server, for example the ISP or router).
3. If the resolver does not know the answer, it asks the root server, then the .in TLD server, and then the authoritative name server of musicmania.in.
4. The authoritative server returns the record (for example A = 192.168.1.50). The resolver caches it and sends it back to the computer, and the browser connects to that IP.

Record	Meaning
SOA	Start of Authority: main information about the zone (primary server, admin e-mail, serial number, timers).
NS	Name server: tells which server is responsible for the zone.
A	Maps a name to an IPv4 address (www > 192.168.1.50).
CNAME	Alias: maps one name to another name (play > www).

Practical: quick step-by-step guide (Kali Linux, BIND9):

1. Install BIND9: sudo apt update, then sudo apt install bind9 bind9utils dnsutils -y. Start it with sudo systemctl enable --now named and check with sudo systemctl status named (it should be active (running); on some systems the service name is bind9).
2. Declare the zone: open sudo nano /etc/bind/named.conf.local and add the zone "musicmania.in" with type master and the file /etc/bind/db.musicmania.in. Save with Ctrl+O, Enter, Ctrl+X.
3. Create the zone file: sudo nano /etc/bind/db.musicmania.in. Write the SOA record, the NS record, the ns1 A record and the www A record (192.168.1.50).
4. Check and restart: sudo named-checkconf, sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in (it must say OK), then sudo systemctl restart named.
5. Test the A record: nslookup www.musicmania.in 127.0.0.1 or dig @127.0.0.1 www.musicmania.in. The answer must be 192.168.1.50.
6. Add the CNAME: open the zone file again, increase the serial number (1 to 2), add the line play IN CNAME www, run named-checkzone and restart named. Then test with nslookup play.musicmania.in 127.0.0.1.
7. If something fails, check the service status, named-checkconf, named-checkzone and the log (sudo journalctl -u named -n 20 --no-pager) as shown in Task 4.

Important points: (1) increase the serial number after every change in the zone file; (2) always write 127.0.0.1 at the end of nslookup (or use dig @127.0.0.1), because musicmania.in is a local test domain and my normal DNS server does not know it; (3) full names in the zone file must end with a dot.

Task 1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

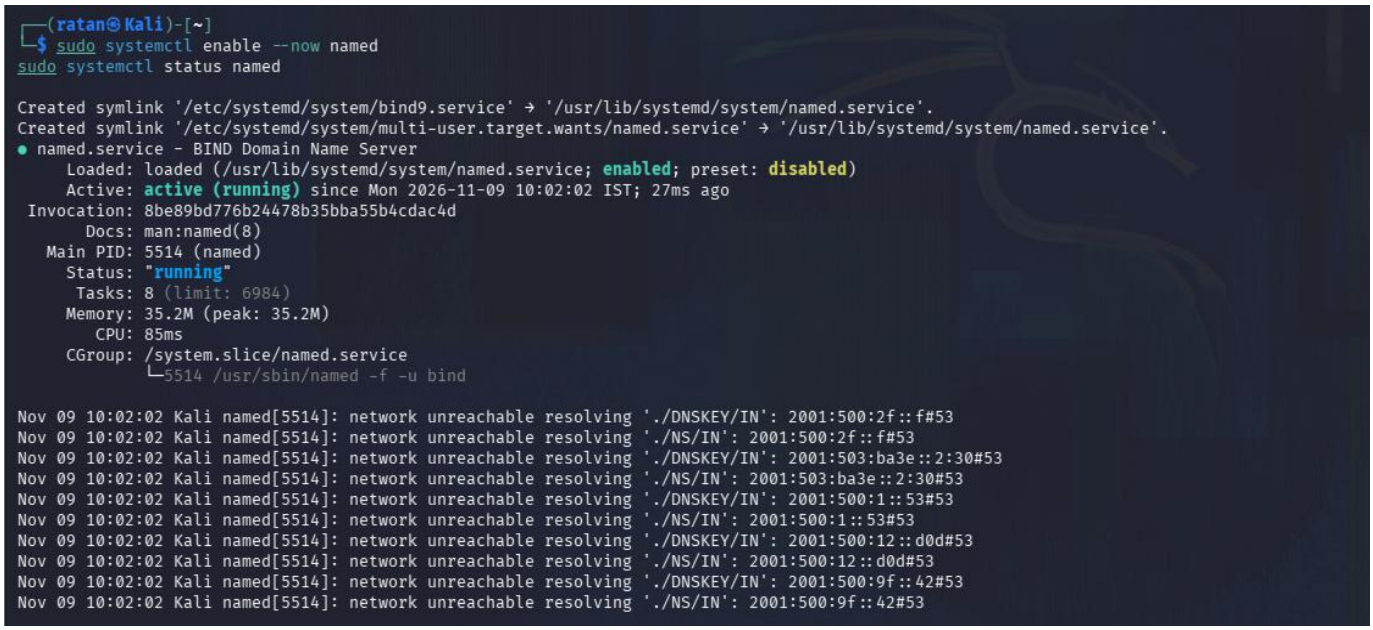
Answer:

I used BIND9 on Kali Linux (a Debian-based Linux).

Step 1: Install BIND9

```
sudo apt update
sudo apt install bind9 bind9utils dnsutils -y
sudo systemctl enable --now named
sudo systemctl status named
```

The service status should show "active (running)".



Step 2: Declare the zone in /etc/bind/named.conf.local

sudo nano /etc/bind/named.conf.local

```
zone "musicmania.in" {
   type master;
   file "/etc/bind/db.musicmania.in";
};
```

Step 3: Create the zone file /etc/bind/db.musicmania.in

sudo nano /etc/bind/db.musicmania.in

```
$TTL   604800
@   IN  SOA ns1.musicmania.in. admin.musicmania.in. (
         1       ; Serial
         604800   ; Refresh
         86400    ; Retry
         2419200  ; Expire
         604800 ) ; Negative Cache TTL
```

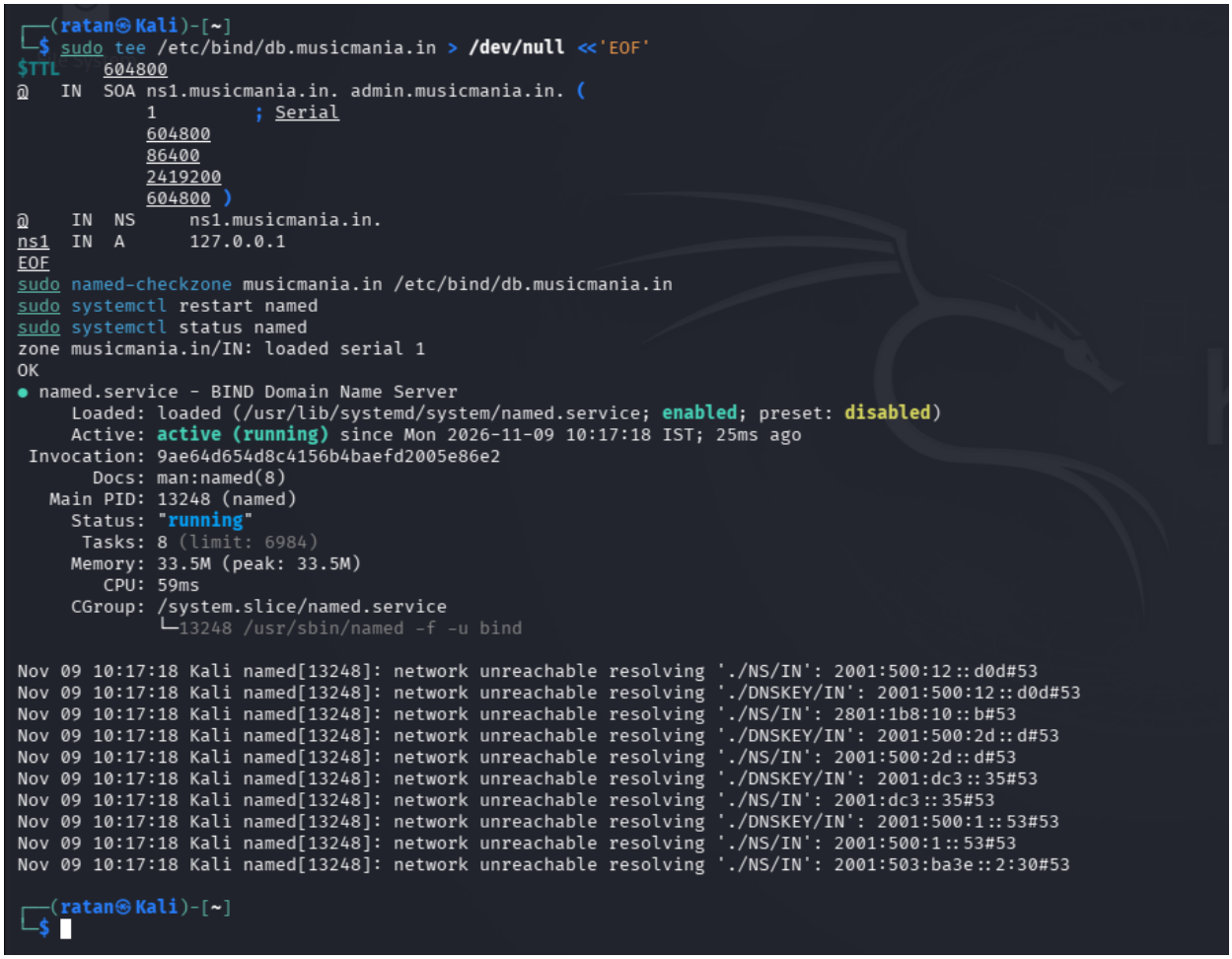
;
@ IN NS ns1.musicmania.in.
ns1 IN A 127.0.0.1

The zone file starts with the SOA record (zone information) and the NS record (the name server of the zone). The name server ns1 points to 127.0.0.1 because I test on the same machine. Every name that ends with a dot (.) is a full name.

Step 4: Check the files and restart the service

```
sudo named-checkconf
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl restart named
```

named-checkconf prints nothing when the main configuration is correct. named-checkzone should print "zone musicmania.in/IN: loaded serial 1" and "OK". The zone musicmania.in is now created.



Task 2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.

Answer:

Step 1: Add the A record at the end of /etc/bind/db.musicmania.in

```
www IN  A    192.168.1.50
```

Step 2: Increase the serial number (1 > 2), check and reload

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl restart named
```

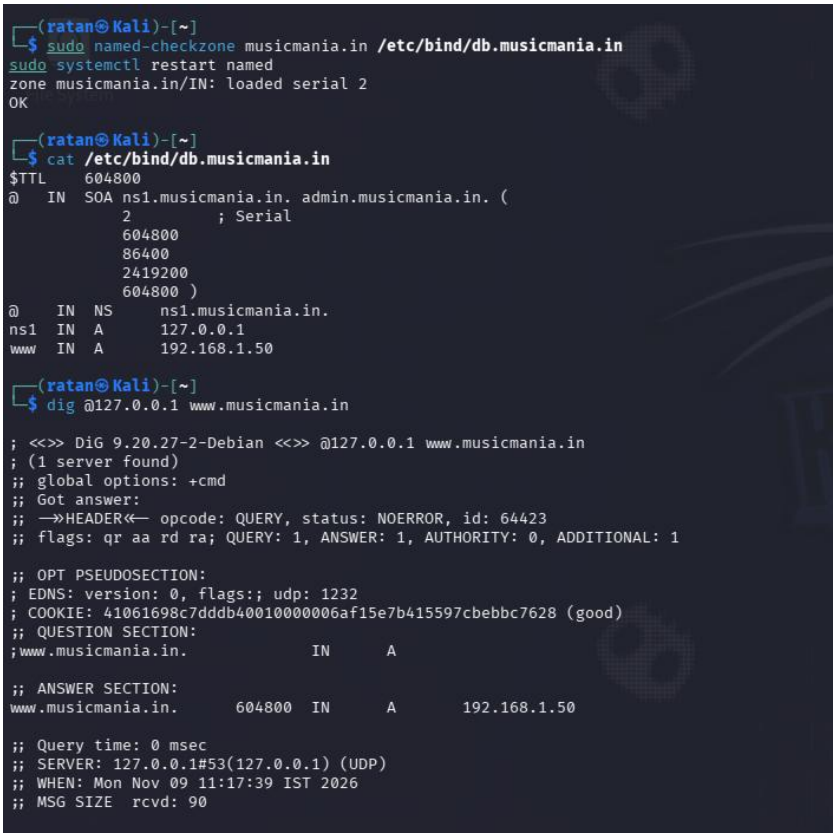
The serial number must be increased every time the zone file is changed, otherwise secondary servers will not notice the change.

Step 3: Test with nslookup and dig (asking my own DNS server)

nslookup www.musicmania.in 127.0.0.1
dig @127.0.0.1 www.musicmania.in A

Expected result when the record is working: the answer section shows www.musicmania.in with the address 192.168.1.50, and dig shows status: NOERROR and flag "aa" (authoritative answer).

Check	Result
Name queried	www.musicmania.in
Address returned	192.168.1.50 (fill from my screenshot)
Server that answered	127.0.0.1#53 (my BIND server)



Task 3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

Answer:

Step 1: Add the CNAME record in the zone file

```
play IN  CNAME www
```

Here "www" is written without a dot, so BIND adds the zone name automatically and it becomes www.musicmania.in. If I wrote "www.musicmania.in" without the final dot by mistake, BIND would make it www.musicmania.in.musicmania.in, which is the typical typo that breaks a CNAME. The correct full form is: play IN CNAME www.musicmania.in. (with the dot at the end).

Step 2: Increase the serial (2 > 3), check and restart

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl restart named
```

Step 3: Test

```
nslookup play.musicmania.in 127.0.0.1
dig @127.0.0.1 play.musicmania.in
```

Expected result: the answer shows that play.musicmania.in is an alias (canonical name) of www.musicmania.in, and then the A record 192.168.1.50. So accessing play.musicmania.in resolves to the same correct IP address as www.

Check	Result
Name queried	play.musicmania.in
Canonical name (CNAME)	www.musicmania.in (fill from my screenshot)
IP address returned	192.168.1.50 (fill from my screenshot)



Final zone file after Tasks 1 to 3:

```
$TTL    604800
@   IN  SOA ns1.musicmania.in. admin.musicmania.in. (
            3         ; Serial
            604800    ; Refresh
            86400     ; Retry
            2419200   ; Expire
            604800 )  ; Negative Cache TTL
;
@    IN  NS     ns1.musicmania.in.
ns1  IN  A      127.0.0.1
www  IN  A      192.168.1.50
play IN  CNAME  www
```

Task 4. Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Answer:

I would go from the simplest check to the deeper ones:

Step 1: Check that the DNS service is running

```
sudo systemctl status named
sudo ss -tulnp | grep :53
```

The service must be active (running) and something must be listening on port 53 (TCP and UDP). If it is stopped, start it with: sudo systemctl start named.

Step 2: Check the configuration file for syntax errors

```
sudo named-checkconf
```

If there is a mistake (for example a missing semicolon or bracket in named.conf.local), this command shows the file name and line number. Also confirm that the zone name is spelled exactly musicmania.in and that the file path in "file" is correct.

Step 3: Check the zone file

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

Common errors here: missing dot at the end of a full name, wrong record type, no NS or SOA record, a CNAME pointing to a name that does not exist, and a wrong IP address. Also check that the serial number was increased after the last change.

Step 4: Restart the service and read the log

```
sudo systemctl restart named
sudo journalctl -u named -n 30 --no-pager
```

The log shows messages like "zone musicmania.in/IN: loaded serial 3" when everything is fine, or the exact error if the zone failed to load. If needed I can also reload only the zone with: sudo rndc reload.

Step 5: Test with nslookup and dig

```
nslookup musicmania.in 127.0.0.1
nslookup www.musicmania.in 127.0.0.1
dig @127.0.0.1 www.musicmania.in +short
dig @127.0.0.1 musicmania.in NS
```

I read the status in the dig output: NOERROR = fine, NXDOMAIN = the name does not exist in the zone (record missing or typo), SERVFAIL = the zone failed to load, REFUSED = the server is not allowed to answer (check allow-query / listen-on). If the result is "connection timed out", the server is not reachable or the firewall blocks port 53.

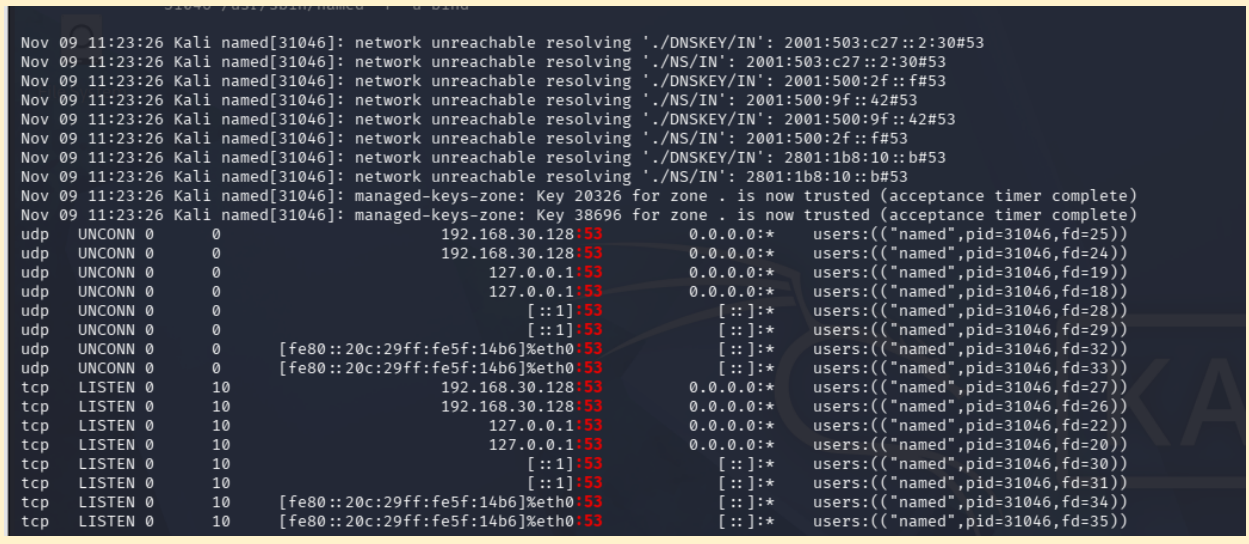
Step 6: Check the client side and the network

```
cat /etc/resolv.conf
ping -c 2 127.0.0.1
sudo ufw status
sudo resolvectl flush-caches
```

The client must use my DNS server in /etc/resolv.conf (nameserver line). If the client still uses another DNS server (for example the router), musicmania.in will never resolve because it is not a real public domain. I also check that the firewall allows port 53, and I clear the DNS cache so that an old answer is not shown.

No.	Problem / symptom	Likely cause and fix
1	Server not answering (timeout)	named is stopped or the firewall blocks port 53 > start the service, allow port 53
2	Service fails to start	Syntax error in named.conf.local > fix the line shown by named-checkconf
3	SERVFAIL	Zone file error > fix it using named-checkzone, then restart
4	NXDOMAIN	Record missing or typo in the name, or missing dot > correct the record and increase the serial
5	Old or wrong IP shown	Serial not increased or cache > increase serial, restart, flush cache
6	Works with @127.0.0.1 but not normally	Client uses a different DNS server > set nameserver in resolv.conf





```
┌──(ratan㉿Kali)-[~]
└─$ sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
zone musicmania.in/IN: loaded serial 3
OK

┌──(ratan㉿Kali)-[~]
└─$ sudo systemctl restart named
sudo journalctl -u named -n 20 --no-pager
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:503:ba3e::2:30#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:7fe::53#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:7fe::53#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:500:12::d0d#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:503:c27::2:30#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:7fd::1#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:7fd::1#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:500:1::53#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:500:1::53#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10::b#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2001:500:2f::f#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './DNSKEY/IN': 2001:500:2f::f#53
Nov 09 11:28:12 Kali named[33442]: network unreachable resolving './NS/IN': 2801:1b8:10::b#53
```

```
┌──(ratan㉿Kali)-[~]
└─$ nslookup musicmania.in 127.0.0.1
nslookup www.musicmania.in 127.0.0.1
dig @127.0.0.1 www.musicmania.in +short
Server:         127.0.0.1
Address:        127.0.0.1#53

** Can't find musicmania.in: No answer

Server:         127.0.0.1
Address:        127.0.0.1#53

Name:   www.musicmania.in
Address: 192.168.1.50

192.168.1.50
```